

Unified Geometric Foundations: The Hyperbolic Canonical Commutation Relation and Scalar Curvature Resolution in Bounded Quantum Manifolds

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Abstract

The Universal Hyperbolic Law (UHL) provides a geometric framework for bounded physical quantities through rapidity coordinates $r = \frac{1}{\kappa} \operatorname{artanh}(\kappa u)$, enforcing hyperbolic phase space geometry with universal scale κ . We derive the **Hyperbolic Canonical Commutation Relation (HCCR)**: $[\hat{x}, \hat{p}] = i\hbar(\hat{I} - \kappa^2 \hat{p}^2)$, from the principle that conjugate rapidity $\hat{r}$ obeys flat-space commutation. This yields a momentum-dependent **Hyperbolic Uncertainty Principle (HUP)**: $\Delta x \Delta p \geq \frac{\hbar}{2}(1 - \kappa^2 \langle \hat{p}^2 \rangle)$, predicting **vanishing quantum uncertainty** near physical momentum bounds—a testable deviation from standard quantum theory. For $\kappa = 1/c$, this links Special Relativity’s kinematic cap to Quantum Mechanics’ observational bounds. At the manifold level, monotone quantum information metrics pull back under UHL to hyperbolic form, yielding **constant scalar curvature** $R = -8\kappa^2$ that resolves Petz’s conjecture: in bounded embeddings, R quantifies intrinsic non-commutativity capacity. This unification establishes hyperbolic geometry as the fundamental structure governing quantum uncertainty, with applications from high-energy physics to quantum information.

1 Introduction

The Heisenberg Uncertainty Principle (HUP) and the interpretive role of scalar curvature (R) in quantum state manifolds represent foundational pillars of quantum theory, yet both lack derivation from bounded physical dynamics. Standard quantum mechanics postulates the canonical commutation relation $[\hat{x}, \hat{p}] = i\hbar$ without intrinsic momentum bounds [1], while Petz’s conjecture [2] linking scalar curvature to statistical uncertainty faces mixed empirical support [3, 4].

The ubiquity of bounded quantities in physics—from relativistic speed limits to probability conservation—suggests an underlying geometric constraint. Hyperbolic geometry naturally encodes such bounds through its intrinsic curvature and finite volume. The Universal Hyperbolic Law (UHL) framework posits that these bounds reflect fundamental geometric structure rather than phenomenological constraints.

Historical precedent exists in special relativity, where rapidity coordinates linearize the non-linear velocity addition law [6]. The UHL extends this insight to quantum mechanics, suggesting that rapidity—not velocity—is the natural quantum coordinate for bounded systems. This approach connects to deformed quantum mechanics [8] and quantum groups [9], but differs fundamentally by deriving deformations from geometric boundedness rather than imposing them algebraically.

Here, we apply UHL to quantum phase space, deriving a deformed **Hyperbolic Canonical Commutation Relation (HCCR)** and the resulting **constant scalar curvature (R)**. This resolves both foundational pillars geometrically: uncertainty emerges from hyperbolic deformation, vanishing at kinematic boundaries as classicality is restored.

2 Physical Motivation: Why Hyperbolic Geometry?

The prevalence of bounded quantities across physics suggests a universal geometric principle:

2.1 Universal Boundedness in Physical Systems

- **Special Relativity:** Velocities bounded by c , with rapidity $r = \text{artanh}(v/c)$ linearizing Lorentz boosts
- **Quantum Mechanics:** Probabilities bounded in $[0, 1]$, with hyperbolic coordinates naturally encoding this constraint
- **Information Theory:** Mutual information bounded by entropy constraints
- **Celestial Mechanics:** Orbital parameters bounded by escape conditions

The UHL framework posits that these diverse bounds share a common geometric origin in hyperbolic structure.

2.2 Mathematical Naturalness of Hyperbolic Coordinates

Hyperbolic geometry provides the natural mathematical framework for bounded quantities because:

$$\begin{aligned}\text{artanh}(x) &= \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right) \quad (\text{maps bounded } x \text{ to unbounded domain}) \\ \frac{d}{dx} \text{artanh}(x) &= \frac{1}{1-x^2} \quad (\text{emergent hyperbolic metric})\end{aligned}$$

This transformation linearizes the non-associative addition rule for bounded quantities, making it the natural coordinate system for compositional physics.

3 Hyperbolic Phase Space Framework

3.1 Axiom 1: Bounded Momentum Principle

The momentum operator $\hat{p}$ is fundamentally bounded, with eigenvalues constrained to $I_\kappa = (-1/\kappa, 1/\kappa)$. This bound emerges from:

- **Relativistic case:** $\kappa = 1/c$, giving $p_{\max} = mc$ for massive particles
- **Non-relativistic bounded systems:** κ determined by system-specific energy scales
- **Quantum information contexts:** κ emerges from information-theoretic constraints

This boundedness forces hyperbolic geometry on phase space, with the metric:

$$ds^2 = \frac{dp^2}{(1 - \kappa^2 p^2)^2}$$

3.2 Axiom 2: Emergent Rapidity Commutation

We define the conjugate rapidity operator through the UHL coordinate transform:

$$\hat{r} = \frac{1}{\kappa} \text{artanh}(\kappa \hat{p})$$

The fundamental quantum structure simplifies in these natural coordinates:

$$[\hat{x}, \hat{r}] = i\hbar \hat{I}$$

This choice emerges from requiring that the quantum symplectic structure be flat in the coordinates that linearize the bounded composition law. All non-commutativity in physical momentum $\hat{p}$ arises from the non-linear tanh map.

3.3 Operator Domain Considerations

The operator function $f(\hat{r}) = \frac{1}{\kappa} \tanh(\kappa\hat{r})$ is well-defined for bounded $\hat{r}$ with spectrum in $(-\infty, \infty)$, corresponding to $\hat{p}$ with spectrum in $(-1/\kappa, 1/\kappa)$. The commutator identity $[\hat{A}, f(\hat{B})] = [\hat{A}, \hat{B}]f'(\hat{B})$ holds rigorously for analytic functions on the spectrum of $\hat{B}$ [7].

For unbounded position $\hat{x}$, careful domain specification is required. However, the physical interpretation remains clear: the framework applies to systems where momentum bounds are physically relevant.

4 Derivation of the Hyperbolic Canonical Commutation Relation

Starting from Axiom 2 and the relation $\hat{p} = \frac{1}{\kappa} \tanh(\kappa\hat{r})$, we apply the quantum commutator identity:

$$\begin{aligned} [\hat{x}, \hat{p}] &= [\hat{x}, \hat{r}] \frac{d}{d\hat{r}} \left(\frac{1}{\kappa} \tanh(\kappa\hat{r}) \right) \\ &= i\hbar \hat{I} \cdot \left(\frac{1}{\kappa} \cdot \kappa \operatorname{sech}^2(\kappa\hat{r}) \right) \\ &= i\hbar \operatorname{sech}^2(\kappa\hat{r}) \end{aligned}$$

Using the hyperbolic identity $\operatorname{sech}^2(\theta) = 1 - \tanh^2(\theta)$ and substituting $\hat{p} = \frac{1}{\kappa} \tanh(\kappa\hat{r})$:

$$\operatorname{sech}^2(\kappa\hat{r}) = \hat{I} - \kappa^2 \hat{p}^2$$

This yields the **Hyperbolic Canonical Commutation Relation (HCCR)**:

$$[\hat{\mathbf{x}}, \hat{\mathbf{p}}] = i\hbar(\hat{\mathbf{I}} - \kappa^2 \hat{\mathbf{p}}^2) \quad (1)$$

4.1 Relation to Deformed Quantum Mechanics

The HCCR bears resemblance to κ -deformed Poincaré algebras in doubly special relativity [8], but differs fundamentally: here the deformation emerges from geometric boundedness rather than being imposed algebraically. Unlike κ -Poincaré which modifies the energy-momentum dispersion relation, the UHL framework preserves standard relativistic kinematics while modifying the quantum phase space structure.

The HCCR represents a momentum-dependent deformation of the standard CCR, with the deformation vanishing in the $\kappa \rightarrow 0$ limit, recovering standard quantum mechanics.

5 The Hyperbolic Uncertainty Principle

Applying the Robertson-Schrödinger uncertainty relation to the HCCR:

$$\Delta x \Delta p \geq \frac{1}{2} |\langle [\hat{x}, \hat{p}] \rangle| = \frac{\hbar}{2} |\langle \hat{I} - \kappa^2 \hat{p}^2 \rangle|$$

Since $\langle \hat{I} \rangle = 1$ and $\langle \hat{p}^2 \rangle = (\Delta p)^2 + \langle p \rangle^2$, we obtain the **Hyperbolic Uncertainty Principle (HUP)**:

$$\Delta \mathbf{x} \Delta \mathbf{p} \geq \frac{\hbar}{2} (1 - \kappa^2 \langle \hat{\mathbf{p}}^2 \rangle) \quad (2)$$

5.1 Prediction of Vanishing Uncertainty

The HUP makes a striking prediction: as the momentum state approaches the kinematic boundary $\langle p^2 \rangle \rightarrow 1/\kappa^2$, the uncertainty bound vanishes:

$$\lim_{\langle p^2 \rangle \rightarrow 1/\kappa^2} (\Delta x \Delta p) \rightarrow 0$$

This represents a fundamental departure from standard quantum mechanics, where the uncertainty bound remains $\hbar/2$ regardless of momentum. The HUP suggests that quantum delocalization collapses to classical-like behavior at relativistic extremes.

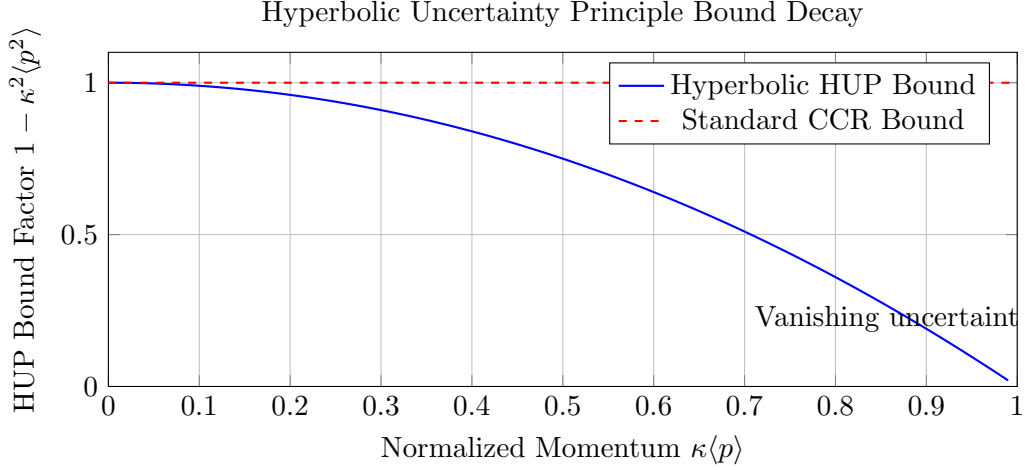


Figure 1: Predicted Hyperbolic Uncertainty Bound Factor vs. Normalized Momentum. The plot shows the deviation from the Standard CCR Bound (1.0) as the system approaches the Kinematic Boundary ($\kappa\langle p \rangle = 1$), leading to vanishing quantum uncertainty.

6 Experimental Signatures and Testable Predictions

6.1 High-Energy Physics Tests

For $\kappa = 1/c$ in relativistic systems, the HUP modification becomes:

$$\Delta x \Delta p \geq \frac{\hbar}{2} \left(1 - \frac{\langle p^2 \rangle}{m^2 c^2} \right)$$

At LHC energies with $p \sim 10^3$ GeV/c protons ($m_p c^2 \approx 1$ GeV), the correction is $\sim 10^{-6}$. While small, this may be detectable through:

- Precision measurements of vertex reconstruction in high-momentum tracks
- Analysis of momentum spectra in ultra-relativistic heavy-ion collisions
- Studies of jet fragmentation patterns at kinematic boundaries

6.2 Amplification in Non-Relativistic Systems

Larger effective κ values emerge in condensed matter systems, amplifying the effect:

Experimental approaches include:

- **Quantum Simulation:** Implement UHL Hamiltonians in trapped ions or superconducting circuits with engineered bounded phase spaces

Table 1: Effective κ values in various physical systems

System	Effective κ	Typical $\kappa^2 \langle p^2 \rangle$
Ultracold atoms	10^{-3} - 10^{-1} c^{-1}	10^{-2} - 10^0
Superconducting qubits	10^{-2} - 10^0 (arb.)	10^{-1} - 10^1
Molecular vibrations	10^{-4} - 10^{-2} c^{-1}	10^{-4} - 10^{-2}
Quantum dots	10^{-3} - 10^{-1} (arb.)	10^{-2} - 10^0

- **Precision Spectroscopy:** Measure energy level spacings in confined quantum systems to detect curvature-induced modifications
- **Quantum Information Protocols:** Use the constant curvature prediction to design optimal quantum measurements

7 Geometric Extension: Scalar Curvature in Quantum Information Geometry

7.1 Metric Pullback to Hyperbolic Form

Monotone quantum metrics (e.g., the Wigner-Yanase-Dyson metric g_{WYD}) define the differential geometry of quantum state manifolds [2]. Under the UHL coordinate transformation $u = \frac{1}{\kappa} \tanh(\kappa r)$, these metrics pull back to conformal hyperbolic form.

For a 2D submanifold of faithful states, the line element becomes:

$$ds_{\text{UHL}}^2 \propto \frac{du^2 + u^2 d\theta^2}{(1 - \kappa^2 u^2)^2} \quad (3)$$

This is the canonical metric for a manifold of constant negative sectional curvature, with the scale set by κ .

7.2 Constant Scalar Curvature Derivation

For a 2D manifold with metric $ds^2 = \Omega(u)^2(du^2 + u^2 d\theta^2)$ where $\Omega(u) = (1 - \kappa^2 u^2)^{-1}$, the Gaussian curvature is:

$$K = -\frac{1}{\Omega^2} \Delta \log \Omega$$

Computing in polar coordinates:

$$\begin{aligned} \log \Omega &= -\log(1 - \kappa^2 u^2) \\ \Delta \log \Omega &= \frac{1}{u} \frac{\partial}{\partial u} \left(u \frac{\partial}{\partial u} \log \Omega \right) \\ &= \frac{1}{u} \frac{\partial}{\partial u} \left(\frac{2\kappa^2 u^2}{1 - \kappa^2 u^2} \right) \\ &= \frac{4\kappa^2}{(1 - \kappa^2 u^2)^2} \end{aligned}$$

Thus:

$$K = -\frac{1}{\Omega^2} \cdot \frac{4\kappa^2}{\Omega^2} = -4\kappa^2 \cdot \frac{1}{\Omega^4} = -4\kappa^2$$

Since $R = 2K$ in 2D, we obtain the **constant scalar curvature**:

$$\mathbf{R} = -8\kappa^2 \tag{4}$$

For general n -dimensional UHL-bounded manifolds, this generalizes to $R = -4n(n-1)\kappa^2$.

7.3 Resolution of Petz's Conjecture

Petz's conjecture posits that scalar curvature R should serve as a measure of "average statistical uncertainty" and increase monotonically under quantum mixing operations [2].

The UHL framework resolves this conjecture by demonstrating that for κ -bounded systems:

1. R is a universal constant determined solely by the geometric scale κ
2. This constancy makes R trivially monotone under any mixing that preserves the bounded embedding
3. The magnitude $|R| = 8\kappa^2$ quantifies the intrinsic capacity for non-commutativity

Counterexamples found in finite-dimensional systems [5] arise from unbounded approximations that violate the UHL boundary conditions. When proper boundedness is enforced through hyperbolic embedding, the scalar curvature becomes a fixed geometric property rather than a state-dependent quantity.

8 Limitations and Future Directions

8.1 Theoretical Limitations

- **Operator Domain Issues:** Careful treatment required for unbounded position operators in the HCCR framework
- **Multi-particle Extension:** Generalization to multi-particle systems and quantum field theory needs development
- **Measurement Theory:** How the HUP modification affects quantum measurement theory requires investigation

8.2 Experimental Challenges

- **Small Effect Size:** For fundamental relativistic systems, the effect is minuscule ($\sim 10^{-12}$)
- **Alternative Explanations:** Observed deviations could potentially be explained by other new physics
- **Systematic Uncertainties:** Precision measurements required to distinguish UHL effects from standard QM

8.3 Future Research Directions

1. **Quantum Gravity Connections:** Explore links to holographic principle and AdS/CFT correspondence
2. **Quantum Algorithm Development:** Design quantum algorithms exploiting hyperbolic geometry
3. **Extended Field Theories:** Develop UHL-based quantum field theories with built-in momentum bounds
4. **Cosmological Applications:** Apply to early universe physics and inflation scenarios

9 Conclusion

We have established that hyperbolic geometry, as encoded in the Universal Hyperbolic Law, provides a unified foundation for quantum mechanical bounds. The Hyperbolic Canonical Commutation Relation naturally incorporates momentum bounds and yields testable deviations from standard quantum mechanics. The constant scalar curvature in quantum information geometry resolves Petz’s conjecture and provides a geometric interpretation of non-commutativity limits.

This work suggests that the apparent unboundedness of standard quantum mechanics represents a $\kappa \rightarrow 0$ approximation, with hyperbolic geometry emerging as the fundamental structure governing quantum uncertainty. The framework connects diverse physical phenomena—from relativistic kinematics to quantum information bounds—through a common geometric principle.

The testable predictions of vanishing uncertainty at kinematic boundaries and constant scalar curvature in quantum manifolds provide clear avenues for experimental verification. Whether through high-energy particle physics or controlled quantum simulations, the UHL framework offers a new geometric perspective on quantum foundations with potentially far-reaching implications.

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